Texas Department of Transportation

Chemical Admixtures for Concrete

Material Producer List

Materials and Tests Division

Chemical Admixtures for concrete

The following materials are qualified in accordance with (Departmental Material Specifications) DMS-4640, "Chemical Admixtures for Concrete."

The dosage rates indicated are the manufacturer's published recommended dosage rates for use under normal conditions. Determine actual dosage rates with trial mixes made with project materials under anticipated construction temperatures and conditions.

1. When using any of the combinations of "Group I, Chemical Admixtures" and "Group II, Air Entraining Admixtures," ensure they are added to the concrete mix separately.
2. Concrete admixtures may exhibit substantially different performances with different brands or changing chemistry of cement. This is usually most noticeable in the retarding effect at a specified dosage. In those cases where an approved admixture does not seem to provide satisfactory performance, please immediately notify the Rigid Pavements and Concrete Materials Section of the Materials and Tests Division at (512) 506-5858.

The Department reserves the right to conduct random sampling of pre-qualified materials for testing and to perform random audits of test reports. Department representatives may sample material from the manufacturing plant, terminal, shipping container, and concrete plant. Material and Tests Division (MTD) reserves the right to test samples to verify compliance with DMS-4640.

Information on the solids content is included when performing a concrete mix-design.

For more information, contact the Materials & Tests Division's Rigid Pavements and Concrete Materials Section at (512) 839-7743.

Group 1, Chemical Admixtures

Type A: Water Reducing Admixtures

- These admixtures reduce the quantity of mixing water required to produce concrete of a given consistency.

Table 1: Approved water Reducing Admixtures (Type A)

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Chryso	98128	Adva 140M	390 to 1300 (6 to 20)	16
Chryso	98128	Adva 190	130 to 325 (2 to 5)	31
Chryso	98128	Adva 405	325 to 650 (5 to 10)	19
Chryso	98128	Chryso EnviroMix 159	195 to 780 (3 to 12)	33
Chryso	98128	Chryso EnviroMix 300	195 to 780 (3 to 12)	33
Chryso	98128	Chryso EnviroMix 330	195 to 652 (3 to 10)	17
Chryso	98128	Chryso EnviroMix 350	195 to 685 (3 to 10)	16
Chryso	98128	Chryso EnviroMix 728	325 to 1170 (5 to 18)	21
Chryso	98128	Chryso EnviroMix 740	195 to 782 (3 to 12)	22
Chryso	98128	Chryso EnviroMix 740i	195 to 782 (3 to 12)	22
Chryso	98128	Chryso EnviroMix i40	195 to 780 (3 to 12)	34
Chryso	98128	Chryso EnviroMix i44	325 to 1170 (5 to 18)	21
Chryso	98128	Chryso Fluid Optima 203	196 to 522 (3 to 8)	22
Chryso	98128	Chryso Fluid Optima 216	522 to 1304 (8 to 20)	22

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Chryso	98128	Chryso Fluid Optima 256 EMx	130 to 587 (2 to 9)	36
Chryso	98128	Chryso Fluid Optima 258 EMx	130 to 457 (2 to 7)	32
Chryso	98128	Chryso Fluid Premia 180	196 to 522 (3 to 8)	25
Chryso	98128	Chryso Fluid Premia 198	65 to 391 (1 to 6)	31
Chryso	98128	Chryso Omega 410	195 to 652 (3 to 10)	18
Chryso	98128	Chryso Optima 223	195 to 522 (3 to 8)	17
Chryso	98128	Chryso Optima 233	130 to 456 (2 to 7)	30
Chryso	98128	Chryso Optima 249	130 to 326 (2 to 5)	33
Chryso	98128	Chryso Optima 1130	130 to 326 (2 to 5)	35
Chryso	98128	Chryso Optima 1240	130 to 391 (2 to 6)	30
Chryso	98128	Chryso Pave 259 EMX	195 to 780 (3 to 12)	33
Chryso	98128	Chryso Plast 209	195 to 390 (3 to 6)	34
Chryso	98128	Chryso Plast 375 EMX	195 to 652 (3 to 10)	32
Chryso	98128	Chryso Plast 405	195 to 390 (3 to 6)	29
Chryso	98128	Chryso Plast CP 40 EMX	195 to 780 (3 to 12)	33
Chryso	98128	Chryso Plast Omega 166	261 to 652 (4 to 10)	18
Chryso	98128	Chryso Plast P140	195 to 520 (3 to 8)	19
Chryso	98128	Chryso Plast P150	195 to 520 (3 to 8)	35

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Chryso	98128	Chryso Plast P400	195 to 390 (3 to 6)	40
Chryso	98128	Chryso PREMIA 576	130 to 587 (2 to 9)	35
Chryso	98128	Chryso Quad 520 EMx	195 to 978 (3 to 15)	24
Chryso	98128	Chryso Quad 822	130 to 652 (2 to 10)	20
Chryso	98128	Chryso Quad 842	130 to 652 (3 to 10)	21
Chryso	98128	Chryso Quad 855 EMx	130 to 652 (2 to 10)	25
Chryso	98128	Chryso Quad 940	130 to 525 (2 to 8)	32
Chryso	98128	Chryso Quad TC 855 EMx	130 to 652 (2 to 10)	25
Chryso	98128	Chryso Rapid Flo i46 EMx	195 to 585 (3 to 9)	19
Chryso	98128	Clarena MC2000	130 to 460 (2 to 7)	16
DarCole Products, Inc.	98387	DNL 485	260 to 650 (4 to 10)	6
DarCole Products, Inc.	98387	DNL 785	390 to 780 (6 to 12)	9
DarCole Products, Inc.	98387	DWR 385	195 to 390 (3 to 6)	34
MAPEI	99289	Dynamon Easy 75	100 to 975 (1.5 to 15)	21
MAPEI	99289	Dynamon EW 100	13 to 975 (2 to 15)	43
MAPEI	99289	Dynamon NRG 546	195 to 975 (3 to 15)	25
MAPEI	99289	Dynamon NRG 1092	195 to 975 (3 to 15)	40
MAPEI	99289	Dynamon SX	200 to 975 (3 to 15)	26

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
MAPEI	99289	Dynamon SX 37	65 to 162 (1 to 2.5)	38
Euclid Chemical Corp.	99924	Eucon A+	195 to 522 (3 to 8)	42
Euclid Chemical Corp.	99924	Eucon Eco Strength	200 to 1040 (3 to 16)	53
Euclid Chemical Corp.	99924	Eucon HC	130 to 390 (2 to 6)	39
Euclid Chemical Corp.	99924	Eucon LR[1]	195 to 260 (3 to 4)	40
Euclid Chemical Corp.	99924	Eucon LW	195 to 390 (3 to 6)	42
Euclid Chemical Corp.	99924	Eucon MR	260 to 910 (4 to 14)	44
Euclid Chemical Corp.	99924	Eucon MRX	195 to 782 (3 to 12)	30
Euclid Chemical Corp.	99924	Eucon NR	195 to 390 (3 to 6)	43
Euclid Chemical Corp.	99924	Eucon NW	195 to 390 (3 to 6)	43
Euclid Chemical Corp.	99924	Eucon SE	130 to 325 (2 to 5)	23
Euclid Chemical Corp.	99924	Eucon SPJ	260 to 455 (4 to 7)	34
Euclid Chemical Corp.	99924	Eucon WR	130 to 650 (2 to 10)	38
Euclid Chemical Corp.	99924	Eucon WR 75	130 to 195 (2 to 3)	44
Euclid Chemical Corp.	99924	Eucon WR 91	130 to 375 (2 to 6)	42
Euclid Chemical Corp.	99924	Eucon X-15	195 to 975 (3 to 15)	43
Euclid Chemical Corp.	99924	Eucon X-20	195 to 975 (3 to 15)	27
Riteks	98250	EvoTek 7015	195 to 1170 (3 to 18)	30

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Riteks	98250	EvoTek 2000	65 to 520 (1 to 8)	40
Riteks	98250	EvoTek PM2020	195 to 975 (3 to 15)	23
Riteks	98250	FloTek 1225	225 to 325 (3.5 to 5)	35
RussTech	99379	ExtendFlo X90	130 to 783 (2 to 12)	28
RussTech	99379	FinishEaseNC	196 to 457 (3 to 7)	44
RussTech	99379	LC400P	196 to 327 (3 to 5)	38
RussTech	99379	LC-500	196 to 327 (3 to 5)	39
Cemex Admix USA	98358	Isocast 6200	130 to 1300 (2 to 20)	41
Cemex Admix USA	98358	Isocast 6300	130 to 1300 (2 to 20)	39
Cemex Admix USA	98358	Isoflex 3650	130 to 975 (2 to 15)	44
Cemex Admix USA	98358	Isoflex 7350	65 to 650 (1 to 10)	19
Cemex Admix USA	98358	Isoflex 8600	65 to 975 (1 to 15)	15
Cemex Admix USA	98358	Isoflow 7730	130 to 975 (2 to 15)	30
Cemex Admix USA	98358	Isoflow 8530	130 to 780 (2 to 12)	30
Cemex Admix USA	98358	Isoflow 9100	130 to 390 (2 to 6)	30
Cemex Admix USA	98358	Isoflow 9750	130 to 520 (2 to 8)	30
Cemex Admix USA	98358	Isoplast 1440	130 to 780 (2 to 12)	38
Cemex Admix USA	98358	Isoplast 3600	65 to 1300 (1 to 20)	42

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
MAPEI Corp.	99289	Mapecube 60	325 to 650 (5 to 10)	24
MAPEI Corp.	99289	Mapecube 60W	200 to 975 (3 to 15)	24
MAPEI Corp.	99289	Mapefluid 441	390 to 1170 (6 to 18)	40
MAPEI Corp.	99289	Mapefluid N200	200 to 1630 (3 to 25)	41
MAPEI Corp.	99289	Mapeplast 400 NC	195 to 455 (3 to 7)	42
MAPEI Corp.	99289	Mapeplast 440 NS	130 to 520 (2 to 8)	15
MAPEI Corp.	99289	Mapeplast KB 1200	195 to 780 (3 to 12)	46
MAPEI Corp.	99289	Mapeplast N	130 to 520 (2 to 8)	38
MAPEI Corp.	99289	Mapeplast Paver Plus	130 to 520 (2 to 88)	38
MAPEI Corp.	99289	MasterEase 5000	65 to 325 (1 to 5)	49
MAPEI Corp.	99289	MasterGlenium 1466	130 to 260 (2 to 4)	40
MAPEI Corp.	99289	MasterGlenium 3030	390 to 1170 (6 to 18)	30
MAPEI Corp.	99289	MasterGlenium 3400	130 to 780 (2 to 12)	44
MAPEI Corp.	99289	MasterGlenium 7500	130 to 975 (2 to 15)	26
MAPEI Corp.	99289	MasterGlenium 7920	130 to 780 (2 to 12)	33
MAPEI Corp.	99289	MasterGlenium 7925	130 to 975 (2 to 15)	30
MAPEI Corp.	99289	MasterGlenium 7960	130 to 780 (2 to 12)	36
MAPEI Corp.	99289	MasterPolyHeed 900	195 to 975 (3 to 15)	41

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
MAPEI Corp.	99289	MasterPolyHeed 980	130 to 520 (2 to 8)	25
MAPEI Corp.	99289	MasterPolyHeed 997	195 to 780 (3 to 15)	48
MAPEI Corp.	99289	MasterPolyHeed 1020	195 to 780 (3 to 12)	24
MAPEI Corp.	99289	MasterPolyHeed 1025	195 to 780 (3 to 12)	25
MAPEI Corp.	99289	MasterPolyHeed 1720	195 to 780 (3 to 12)	19
MAPEI Corp.	99289	MasterPolyHeed 1725	195 to 780 (3 to 12)	18
MAPEI Corp.	99289	MasterPolyHeed 1855	130 to 520 (2 to 8)	15
MAPEI Corp.	99289	MasterPozzolith 80	260 to 650 (4 to 10)	41
MAPEI Corp.	99289	MasterPozzolith 200	195 to 325 (3 to 5)	32
MAPEI Corp.	99289	MasterPozzolith 322	195 to 455 (3 to 7)	44
MAPEI Corp.	99289	MasterPozzolith 700	195 to 325 (3 to 5)	28
MAPEI Corp.	99289	Master X-Seed 66	260 to 975 (4 to 15)	38
Chryso	98128	MIRA 62	130 to 390 (2 to 6)	18
Chryso	98128	MIRA 95	130 to 460 (2 to 7)	11
Chryso	98128	MIRA 110	195 to 325 (3 to 5)	17
Riteks	98250	NexCrete 1200	196 to 780 (3 to 12)	45
Riteks	98250	NexCrete 1410	65 to 391 (1 to 6)	20
Riteks	98250	NexCrete 1610	65 to 1304 (1 to 20)	22

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Riteks	98250	NexCrete 1L-X	65 to 520 (1 to 8)	19
Riteks	98250	NexCrete ES	65 to 1300 (1 to 20)	22
Riteks	98250	NexCrete HTX	196 to 1697 (3 to 26)	30
Euclid Chemical Corp.	99924	Plastol 341	130 to 650 (2 to 10)	17
Euclid Chemical Corp.	99924	Plastol 6420	130 to 650 (2 to 10)	16
Euclid Chemical Corp.	99924	Plastol 6425	200 to 1000 (3 to 15)	31
Euclid Chemical Corp.	99924	Plastol SPC	195 to 390 (3 to 6)	41
Euclid Chemical Corp.	99924	Plastol Ultra 109	130 to 780 (2 to 12)	40
MAPEI Corp.	99289	Polychem SPC	130 to 190 (2 to 3)	37
Sika Corp	99572	SikaControl-312 SE	130 to 1045 (2 to 16)	42
Sika Corp	99572	Sika Plastocrete 161	195 to 325 (3 to 8)	28
Sika Corp	99572	Sika Plastocrete 250	130 to 590 (2 to 9)	16
Sika Corp	99572	Sika ViscoCrete 1000	130 to 1170 (2 to 18)	30
Sika Corp	99572	Sika ViscoCrete 1100	65 to 780 (1 to 12)	46
Sika Corp	99572	Sika ViscoCrete 2100	130 to 1040 (2 to 16)	40
Sika Corp	99572	Sika ViscoCrete 4100	195 to 780 (3 to 12)	40
Sika Corp	99572	Sika ViscoCrete 5100	65 to 780 (1 to 12)	41
Sika Corp	99572	Sika ViscoCrete 6100	195 to 1040 (3 to 16)	40

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Sika Corp	99572	Sika ViscoFlow-2020	130 to 520 (2 to 8)	21
Sika Corp	99572	Sika ViscoFlow 585	195 to 1400 (3 to 20)	25
Sika Corp	99572	Sikament 686	195 to 780 (3 to 12)	20
Sika Corp	99572	Sikament AFM	195 to 780 (3 to 12)	15
Sika Corp	99572	Sikament N	390 to 1300 (6 to 20)	41
Sika Corp	99572	Sikament 475	200 to 975 (3 to 15)	20
Sika Corp	99572	SikaPlast 300 GP	130 to 780 (2 to 12)	19
Sika Corp	99572	SikaPlast 400 LF	130 to 780 (2 to 12)	25
Sika Corp	99572	SikaPlast 500	195 to 780 (3 to 12)	33
Riteks	98250	Slump Retain 240	130 to 780 (2 to 12)	25
RussTech	99379	Superflow 2000 SCC	196 to 1697 (3 to 26)	30
RussTech	99379	Superflow 2000 RM	65 to 391 (1 to 6)	20
RussTech	99379	Superflow 2040 RM	65 to 522 (1 to 8)	39
Chryso	98128	WRDA 64	195 to 390 (3 to 6)	44
Chryso	98128	ZYLA 610	130 to 260 (2 to 4)	21
Chryso	98128	ZYLA 630	130 to 455 (2 to 7)	23
Chryso	98128	ZYLA 640	130 to 455 (2 to 7)	19

Type B: Retarding Admixtures

- These admixtures retard the setting of concrete.

Table 2: Approved Retarding Admixtures (Type B)

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Chryso	98128	CHRYSO NutralSet TC	65 to 780 (1 to 12)	17
DarCole Products, Inc	98387	DHS	65 to 780 (1 to 12)	13
DarCole Products, Inc	98387	DWR 385R	65 to 780 (1 to 12)	32
Euclid Chemical Corp.	99924	Eucon DS	65 to 1000 (1 to 16)	26
Euclid Chemical Corp.	99924	Eucon LR	260 to 650 (4 to 10)	40
Euclid Chemical Corp.	99924	Eucon NR	260 to 520 (4 to 8)	43
Euclid Chemical Corp.	99924	Eucon Retarder 75	195 to 325 (3 to 5)	34
Euclid Chemical Corp.	99924	Eucon Retarder 100	130 to 390 (2 to 6)	39
Euclid Chemical Corp.	99924	Eucon Stasis	65 to 1040 (1 to 16)	18
MAPEI Corp.	99289	Mapecrete Resolve	65 to 780 (1 to 12)	14
MAPEI Corp.	99289	Mapetard Plus	130 to 325 (2 to 5)	33
Master Builders Solutions US LLC	99470	MasterPozzolith 200	260 to 390 (4 to 6)	34
Master Builders Solutions US LLC	99470	MasterPozzolith 700	195 to 450 (3 to 7)	28
Master Builders Solutions US LLC	99470	MasterSet R 961	195 to 325 (3 to 5)	34
MAPEI Corp.	99289	Polychem Renu	65 to 780 (1 to 12)	14

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Sika Corp.	99572	Sika Plastiment XR	130 to 325 (2 to 5)	15
Sika Corp.	99572	SikaTard 440	130 to 520 (2 to 8)	20
Riteks	98250	SYAD 40	33 to 390 (0.5 to 6)	35
Riteks	98250	SYAD 75	100 to 195 (1.5 to 3)	25

Type C: Accelerating Admixtures (Non-Chloride)

- These admixtures accelerate the setting and early-strength development of concrete.
- Do not use accelerating admixtures on bridge decks at the top slab of direct traffic culverts.

Table 3: Approved Retarding Admixtures (Type C)

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Euclid Chemical Corp.	99924	Accelguard 80	1040 to 2080 (16 to 32)	60
Euclid Chemical Corp.	99924	Accelguard AcN 200	650 to 3900 (10 to 60)	42
Euclid Chemical Corp.	99924	Accelguard G3	650 to 5870 (10 to 90)	40
Euclid Chemical Corp.	99924	Accelguard NCA	780 to 4875 (12 to 75)	59
Riteks	98250	Advanset NC	522 to 5850 (8 to 90)	45
Riteks	98250	Advanset NC14	650 to 2935 (10 to 45)	14
Riteks	98250	Advanset RCI	522 to 5870 (8 to 90)	31
Chryso	98128	Chryso CI 100	500 to 3500 (7.5 to 54)	31
Chryso	98128	Chryso TurboCast 650A	397 to 5850 (6 to 90)	47
Chryso	98128	Chryso TurboCast 701	650 to 5850 (10 to 90)	34
Chryso	98128	Chryso TurboCast NCT	520 to 5850 (8 to 90)	62
Chryso	98128	Daraset 400	650 to 3910 (10 to 60)	54
Chryso	98128	DCI Corrosion Inhibitor	2730 to 8320 (42 to 128)	33
DarCole Products, Inc.	98387	DNS 130	585 to 3510 (9 to 54)	47

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
DarCole Products, Inc.	98387	DNS 166	585 to 3510 (9 to 54)	54
Euclid Chemical Corp.	99924	Eucon CIA	3250 to 9750 (50 to 150)	32
Cemex Admix USA	98358	Isoxel 5400	650 to 6500 (10 to 100)	56
MAPEI Corp.	99289	Mapefast Super Set	520 to 2080 (8 to 32)	59
MAPEI Corp.	99289	Mapefast Super Set Plus	520 to 2080 (8 to32)	59
MAPEI Corp.	99289	Mapeshield Cl 300	520 to 5850 (8 to 90)	30
Master Builders Solutions US LLC	99470	MasterLife CI 30	2600 to 5850 (40 to 90)	33
Master Builders Solutions US LLC	99470	MasterSet FP 20	325 to 3910 (5 to 60)	48
Master Builders Solutions US LLC	99470	MasterSet AC 534	650 to 2925 (10 to 45)	56
Fritz-Pak Corp.	99803	NCA	1000 to 3000 (16 to 48)	98
Chryso	98218	Polarset	520 to 6500 (8 to 100)	42
RussTech	99379	RussTech RCI	522 to 5870 (8 to 90)	31
Sika Corp.	99572	Sika CNI	3250 to 9750 (50 to 150)	33
Sika Corp.	99572	Sika Mix NCA	650 to 2925 (10 to 45)	46
Sika Corp.	99572	Sika Plastocrete 161FL	390 to 4160 (6 to 64)	43
Sika Corp.	99572	Sika Rapid -1	520 to 3120 (8 to 48)	32
Sika Corp.	99572	Sika Set NC-4	580 to 2600 (9 to 40)	58

Type D: Water – Reducing, Retarding Admixtures

- These admixtures reduce the quantity of mixing water required to produce concrete of a given consistency and retard the setting of concrete.

Table 4: Approved Water Reducing Retarding Admixtures (Type D)

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Chryso	98128	Chryso ControlSet 65	130 to 325 (2 to 5)	32
Chryso	98128	Chryso ControlSet 105	130 to 520 (2 to 8)	20
Chryso	98128	Chryso ControlSet 125	130 to 456 (2 to 7)	17
Chryso	98128	Chryso Plast P400	260 to 455 (4 to 7)	40
Chryso	98128	Chryso Plast 405	260 to 455 (4 to 7)	29
Chryso	98128	Chryso SetControl iTC	130 to 325 (2 to 5)	22
Chryso	98128	Daratard 17	130 to 520 (2 to 8)	50
DarCole Products, Inc	98387	DHS	65 to 780 (1 to 12)	13
DarCole Products, Inc	98387	DWR 385R	65 to 780 (1 to 12)	32
Euclid Chemical Corp.	99924	Eucon HC	130 to 390 (2 to 6)	39
Euclid Chemical Corp.	99924	Eucon LR[1]	260 to 650 (4 to 10)	40
Euclid Chemical Corp.	99924	Eucon NR	260 to 520 (4 to 8)	43
Euclid Chemical Corp.	99924	Eucon NW	130 to 782 (2 to 6)	41
Euclid Chemical Corp.	99924	Eucon Retarder 75	195 to 325 (3 to 5)	34

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Euclid Chemical Corp.	99924	Eucon Retarder 100	130 to 390 (2 to 6)	39
Euclid Chemical Corp.	99924	Eucon SE	130 to 325 (2 to 5)	23
Euclid Chemical Corp.	99924	Eucon Stasis	65 to 1040 (1 to 16)	18
Riteks	98250	FloTek 1225	225 to 325 (3.5 to 5)	35
Riteks	98250	FloTek 1240	325 to 522 (5 to 8)	35
Fritz-Pak Corp.	99803	FR-1	100 to 120 (1.5 to 2)	92
Riteks	98250	HydraStop	65 to 485 (1 to 7.5)	16
Cemex Admix USA	98358	Isoflex 7350	65 to 650 (1 to 10)	19
Cemex Admix USA	98358	Isopause 2370	130 to 520 (2 to 8)	43
Cemex Admix USA	98358	Isopause 2500	65 to 650 (1 to 10)	19
Cemex Admix USA	98358	Isoplast 1440	130 to 780 (2 to 12)	38
RussTech	99379	LC400P	260 to 520 (4 to 8)	38
RussTech	99379	LC400R	65 to 390 (1 to 6)	33
RussTech	99379	LC-500	196 to 327 (3 to 5)	39
MAPEI Corp.	99289	Mapeplast 400 NC	260 to 390 (4 to 6)	41
MAPEI Corp.	99289	Mapeplast 440 NS	130 to 520 (2 to 8)	15
MAPEI Corp.	99289	Mapeplast N	130 to 520 (2 to 8)	38
MAPEI Corp.	99289	Mapecrete Resolve	65 to 780 (1 to 12)	14
MAPEI Corp.	99289	Mapetard R	130 to 325 (2 to 5)	45

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Master Builders Solutions US LLC	99470	MasterPozzolith 80	455 to 650 (7 to 10)	41
MAPEI Corp.	99289	MasterPozzolith 200	260 to 390 (4 to 6)	50
MAPEI Corp.	99289	MasterPozzolith 700	195 to 450 (3 to 7)	28
MAPEI Corp.	99289	MasterSet Delvo	195 to 325 (3 to 5)	14
MAPEI Corp.	99289	Master Set R 99	130 to 260 (2 to 4)	25
MAPEI Corp.	99289	MasterSet R 100	130 to 260 (2 to 4)	50
MAPEI Corp.	99289	MasterSet R 300	195 to 325 (3 to 5)	50
MAPEI Corp.	99289	MasterSet R 961	195 to 325 (3 to 5)	34
MAPEI Corp.	99289	Polychem Renu	65 to 780 (1 to 12)	14
Chryso	98128	Recover	130 to 780 (2 to 12)	21
RussTech	99379	RENU	65 to 783 (1 to 12)	13
Sika Corp.	99572	Sika Plastiment	6 to 390 (0.1 to 6)	33
Sika Corp.	99572	Sika Plastiment XR	6 to 325 (0.1 to 5)	15
Sika Corp.	99572	Sika Plastocrete 250	130 to 590 (2 to 9)	16
Sika Corp.	99572	SikaTard 440	6 to 1320 (0.1 to 48)	20
Sika Corp.	99572	SikaTard 550	6 to 1320 (0.1 to 48)	17
Fritz-Pak Corp.	99803	Standard Delayed Set	200 to 300 (3 to 5)	93
Riteks	98250	SYAD 40	33 to 390 (0.5 to 6)	35

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Chryso	98128	V-Mar VSC500	455 to 975 (7 to 15)	16
Chryso	98128	WRDA 64	260 to 590 (4 to 9)	44
Chryso	98128	WRDA 79	325 to 650 (5 to 10)	39
Chryso	98128	Zyla 610	260 to 783 (4 to 7)	20
Chryso	98128	Zyla 614R	130 to 300 (2 to 5)	28
Chryso	98128	Zyla 630	195 to 445 (5 to 7)	23
Chryso	98128	Zyla R	195 to 325 (3 to 5)	12

Type E: Water – Reducing and Accelerating Admixtures (Non-Chloride)

- These admixtures accelerate the setting and early-strength development of concrete while reducing water content.
- Do not use accelerating admixtures on bridge decks or on top slab of direct traffic culverts.

Table 5: Approved Retarding Admixtures (Type E)

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Euclid Chemical Corp.	99924	Accelguard 80	780 to 5870 (12 to 90)	59
Euclid Chemical Corp.	99924	Accelguard AcN 200	650 to 3910 (10 to 60)	46
Euclid Chemical Corp.	99924	Accelguard G3	650 to 5870 (10 to 90)	40
Euclid Chemical Corp.	99924	Accelguard NCA	780 to 4890 (12 to 75)	58
DarCole Products, Inc.	98387	DNS 166	585 to 3510 (9 to 54)	54
RussTech	99379	LCNC-166	522 to 5870 (8 to 90)	54
Master Builders Solutions US LLC	99470	MasterSet FP 20	325 to 3910 (5 to 60)	48
Fritz-Pak Corp.	99803	NCA	1000 to 3000 (16 to 48)	98
Sika Corp.	99572	Sika Plastocrete 161FL	390 to 4160 (6 to 64)	43
Sika Corp.	99572	Sika Set NC-4	580 to 2600 (9 to 40)	58

Type F: High Range Water-Reducing Admixtures (HRWR)

- These are admixtures that reduce the quantity of mixing water required to produce concrete of a given consistency by 12% or greater.
- Do not use this type of admixture on bridge deck concrete.

Table 6: Approved High Range Water-Reducing Admixtures (Type F)

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Chryso	98128	ADVA 140M	390 to 1300 (6 to 20)	16
Chryso	98128	ADVA 190	195 to 590 (3 to 9)	30
Chryso	98128	ADVA 195	195 to 975 (3 to 15)	32
Chryso	98128	ADVA 198	130 to 975 (2 to 15)	26
Chryso	98128	ADVA 405	650 to 1175 (10 to 18)	19
Chryso	98128	ADVA Cast 555	400 to 1400 (6 to 20)	16
Chryso	98128	ADVA Cast 575	130 to 650 (2 to 10)	40
Chryso	98128	ADVA Cast 585	130 to 650 (2 to 10)	36
Chryso	98128	ADVA Cast 593	130 to 650 (2 to 10)	37
Chryso	98128	ADVA Cast 600	130 to 650 (2 to 10)	38
Chryso	98128	ADVA Flex	260 to 910 (4 to 14)	29
Chryso	98128	ADVA ITM750	195 to 650 (3 to 10)	34
Chryso	98128	Chryso Enviromix 330	456 to 1174 (7 to 18)	17
Chryso	98128	Chryso Enviromix 350	522 to 1174 (8 to 18)	16

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Chryso	98128	Chryso Enviromix 728	652 to 1170 (10 to 18)	17
Chryso	98128	Chryso Enviromix 740	567 to 1304 (9 to 20)	22
Chryso	98128	Chryso Enviromix 740i	567 to 1304 (9 to 20)	22
Chryso	98128	Chryso Enviromix i40	780 to 1304 (12 to 20)	34
Chryso	98128	Chryso Enviromix i44	652 to 1170 (10 to 18)	17
Chryso	98128	Chryso Fluid Optima 216	522 to 1304 (8 to 20)	22
Chryso	98128	Chryso Fluid Optima 256 EMx	457 to 1045 (7 to 16)	36
Chryso	98128	Chryso Fluid Optima 258 EMx	457 to 1045 (7 to 16)	32
Chryso	98128	Chryso Fluid Premia 198	196 to 1304 (3 to 20)	31
Chryso	98128	Chryso Fluid ProPel 3300	520 to 1430 (8 to 22)	40
Chryso	98128	Chryso Fluid ProPel AG	650 to 1300 (10 to 20)	39
Chryso	98128	Chryso Omega 410	391 to 1304 (6 to 20)	18
Chryso	98128	Chryso Optima 223	391 to 1304 (6 to 20)	17
Chryso	98128	Chryso Optima 233	326 to 978 (5 to 15)	30
Chryso	98128	Chryso Optima 249	261 to 782 (4 to 12)	33
Chryso	98128	Chryso Optima 1130	261 to 782 (4 to 12)	36
Chryso	98128	Chryso Optima 1240	261 to 1043 (4 to 16)	30
Chryso	98128	Chryso Quad 822	391 to 1304 (6 to 20)	20

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Chryso	98128	Chryso Quad 842	391 to 1304 (6 to 20)	21
Chryso	98128	Chryso Quad 940	260 to 1304 (4 to 20)	32
Chryso	98128	Chryso Quad TC 855 EMx	391 to 1304 (6 to 20)	25
Chryso	98128	Chryso RapidFlo i46 EMx	652 to 1170 (10 to 18)	19
Chryso	98128	Chryso PREMIA 576	130 to 587 (2 to 9)	35
Chryso	98128	CLARENA MC2000	460 to 975 (7 to 15)	16
Chryso	98128	CONCERA SA8080	520 to 1300 (8 to 20)	20
Chryso	98128	Daracem 19	390 to 1300 (6 to 20)	40
DarCole Products, Inc.	98387	DSF 1443	195 to 1690 (3 to 26)	29
MAPEI Corp.	99289	Dynamon Easy 75	100 to 975 (1.5 to 15)	21
MAPEI Corp.	99289	Dynamon NRG 546	195 to 975 (3 to 15)	25
MAPEI Corp.	99289	Dynamon NRG 1092	195 to 975 (3 to 15)	40
MAPEI Corp.	99289	Dynamon SX	180 to 975 (3 to 15)	26
MAPEI Corp.	99289	Dynamon SX 37	130 to 715 (2 to 11)	38
Euclid Chemical Corp.	99924	Eucon 37	650 to 1040 (10 to 16)	40
Euclid Chemical Corp.	99924	Eucon 1037	650 to 1625 (10 to 25)	41
Euclid Chemical Corp.	99924	Eucon MRX	195 to 782 (3 to 12)	30
Euclid Chemical Corp.	99924	Eucon SP	390 to 1625 (6 to 25)	40

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Euclid Chemical Corp.	99924	Eucon SPJ	260 to 455 (4 to 7)	34
Euclid Chemical Corp.	99924	Eucon X-15	195 to 975 (3 to 15)	43
Euclid Chemical Corp.	99924	Eucon X-20	195 to 975 (3 to 15)	27
Riteks	98250	EvoTek 40X	195 to 975 (3 to 15)	40
Riteks	98250	EvoTek 2000	65 to 520 (1 to 8)	40
Riteks	98250	EvoTek 7015	195 to 1170 (3 to 18)	30
Riteks	98250	EvoTek PM2020	195 to 975 (3 to 15)	23
Chryso	98218	EXP 950	130 to 650 (2 to 10)	40
Cemex Admix USA	98358	Isocast 6200	130 to 1300 (2 to 20)	41
Cemex Admix USA	98358	Isocast 6300	130 to 1300 (2 to 20)	39
Cemex Admix USA	98358	Isoflex 8600	65 to 975 (1 to 15)	15
Cemex Admix USA	98358	Isoflow 7730	130 to 975 (2 to 15)	30
Cemex Admix USA	98358	Isoflow 8530	130 to 780 (2 to 12)	30
Cemex Admix USA	98358	Isoflow 9100	130 to 390 (2 to 6)	30
Cemex Admix USA	98358	Isoflow 9750	65 to 650 (1 to 10)	30
Cemex Admix USA	98358	Isoplast 3600	65 to 1300 (1 to 20)	42
MAPEI Corp.	99289	Mapefluid 441	390 to 1170 (6 to 18)	40
MAPEI Corp.	99289	Mapefluid N200	200 to 1630 (3 to 25)	41

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Master Builders Solutions US LLC	99470	MasterGlenium 1466	260 to 650 (4 to 10)	40
Master Builders Solutions US LLC	99470	MasterGlenium 3030	390 to 1170 (6 to 18)	20
Master Builders Solutions US LLC	99470	MasterGlenium 3400	130 to 780 (2 to 12)	45
Master Builders Solutions US LLC	99470	MasterGlenium 7101	195 to 780 (3 to 12)	32
Master Builders Solutions US LLC	99470	MasterGlenium 7500	130 to 975 (2 to 15)	26
Master Builders Solutions US LLC	99470	MasterGlenium 7700	260 to 975 (4 to 15)	34
Master Builders Solutions US LLC	99470	MasterGlenium 7710	130 to 975 (2 to 15)	25
Master Builders Solutions US LLC	99470	MasterGlenium 7920	130 to 780 (2 to 12)	33
Master Builders Solutions US LLC	99470	MasterGlenium 7925	130 to 975 (2 to 15)	30
Master Builders Solutions US LLC	99470	MasterGlenium 7960	260 to 780 (4 to 12)	36
Master Builders Solutions US LLC	99470	MasterPolyHeed 980	390 to 780 (6 to 12)	25
Master Builders Solutions US LLC	99470	MasterPolyHeed 997	195 to 780 (3 to 15)	48
Master Builders Solutions US LLC	99470	MasterPolyHeed 1020	195 to 780 (3 to 12)	24
Master Builders Solutions US LLC	99470	MasterPolyHeed 1025	195 to 780 (3 to 12)	25
Master Builders Solutions US LLC	99470	MasterPolyHeed 1720	195 to 780 (3 to 12)	19
Master Builders Solutions US LLC	99470	MasterPolyHeed 1725	195 to 780 (3 to 12)	18
Master Builders Solutions US LLC	99470	MasterPolyHeed 1855	520 to 780 (8 to 12)	15
Master Builders Solutions US LLC	99470	MasterRheobuild 1000	650 to 1625 (10 to 25)	41

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Chryso	98128	MIRA 62	390 to 975 (6 to 15)	18
Chryso	98128	MIRA 95	130 to 975 (2 to 15)	11
Chryso	98128	MIRA 110	350 to 975 (6 to 15)	17
Riteks	98250	NexCrete 1410	457 to 1300 (7 to 20)	20
Riteks	98250	NexCrete 1L-X	65 to 520 (1 to 8)	19
Riteks	98250	NexCrete ES	65 to 1300 (1 to 20)	22
Riteks	98250	NexCrete HTX	196 to 1697 (3 to 26)	30
Euclid Chemical Corp.	99924	Plastol 341	130 to 165 (2 to 10)	17
Euclid Chemical Corp.	99924	Plastol 5000	195 to 975 (3 to 15)	32
Euclid Chemical Corp.	99924	Plastol 6400	200 to 780 (3 to 12)	36
Euclid Chemical Corp.	99924	Plastol 6425	200 to 1000 (3 to 15)	31
Euclid Chemical Corp.	99924	Plastol 6420	130 to 650 (2 to 10)	17
Euclid Chemical Corp.	99924	Plastol SPC	195 to 800 (3 to 12)	40
Euclid Chemical Corp.	99924	Plastol Ultra 109	130 to 780 (2 to 12)	40
Euclid Chemical Corp.	99924	Plastol Ultra 209	130 to 780 (2 to 12)	36
MAPEI Corp.	99289	Polychem Paver Plus	190 to 392 (3 to 6)	37
MAPEI Corp.	99289	Polychem SPC	130 to 520 (2 to 8)	38
Hexion Inc.	99536	SAFSP - 101	910 to 2080 (14 to 32)	40

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Sika Corp.	99572	Sika ViscoCrete 1000	130 to 1170 (2 to 18)	30
Sika Corp.	99572	Sika ViscoCrete 1100	65 to 780 (1 to 12)	46
Sika Corp.	99572	Sika ViscoCrete 2100	130 to 1040 (2 to 16)	40
Sika Corp.	99572	Sika ViscoCrete 2110	195 to 780 (3 to 12)	42
Sika Corp.	99572	Sika ViscoCrete 4100	195 to 780 (3 to 12)	40
Sika Corp.	99572	Sika ViscoCrete 5100	65 to 780 (1 to 12)	41
Sika Corp.	99572	Sika ViscoCrete 6100	195 to 1040 (3 to 16)	40
Sika Corp.	99572	Sika ViscoFlow 2020	130 to 520 (2 to 8)	21
Sika Corp.	99572	Sika ViscoFlow 585	195 to 1400 (3 to 20)	25
Sika Corp.	99572	Sikament AFM	390 to 780 (6 to 12)	15
Sika Corp.	99572	Sikament N	390 to 1300 (6 to 20)	41
Sika Corp.	99572	Sikament 475	200 to 975 (3 to 15)	20
Sika Corp.	99572	Sikament 686	195 to 780 (3 to 12)	20
Sika Corp.	99572	SikaMix HC-300	130 to 650 (2 to 10)	24
Sika Corp.	99572	SikaPlast 300GP	130 to 780 (2 to 12)	19
Sika Corp.	99572	SikaPlast 400 LF	130 to 780 (2 to 12)	25
Sika Corp.	99572	SikaPlast 500	195 to 780 (3 to 12)	33
Fritz-Pak Corp.	99803	Supercizer 1	300 to 450 (5 to 7)	95

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Fritz-Pak Corp.	99803	Supercizer 5	300 to 450 (5 to 7)	91
RussTech	99379	Superflow 2000 SCC	196 to 1697 (3 to 26)	30
RussTech	99379	Superflow 2000 RM	457 to 1300 (7 to 20)	20
RussTech	99379	Superflow 2040 RM	65 to 522 (1 to 8)	39

Type G: Water-Reducing High Range Retarding Admixtures

- These are admixtures that reduce the quantity of mixing water required to produce concrete of a given consistency by 12% or greater and retard the setting of concrete.
- Do not use this type of admixture on bridge deck concrete.

Table 7: Approved Retarding Admixtures (Type G)

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Chryso	98128	Chryso Fluid Optima 100	260 to 780 (4 to 30)	30
Euclid Chemical Corp.	99924	Eucon 537	650 to 1300 (10 to 20)	43
RussTech	99379	FinishEase-NC	522 to 783 (8 to 12)	44
Fritz-Pak Corp.	99803	Supercizer 7	250 to 750 (4 to 12)	89

Group II, Air Entraining Admixtures

Air Entraining Admixture

- These are admixtures added to the batch for entraining air.
- The dosage MUST be adjusted as necessary to produce the required air content in the concrete within the specified tolerances.

Table 8: Approved Air Entraining Admixtures

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Fritz-Pak Corp.	99803	Air Plus	16 to 81 (0.25 to 1.25)	98
Chryso	98128	Airalon 3000	30 to 200 (0.5 to 3)	3
Chryso	98128	Airalon 7000	30 to 200 (0.5 to 3)	7
Riteks	98250	A-Train FF	33 to 130 (0.5 to 2)	7
Riteks	98250	A-Train PR	33 to 130 (0.5 to 2)	10
Riteks	98250	A-Train VR	16 to 260 (0.25 to 4)	10
Cellular Concrete Technologies	98274	CCT Stable Air	65 to 580 (1 to 8.9)	31
Chryso	98128	Chryso Air 260	13 to 195 (0.2 to 3)	7
Chryso	98128	Chryso Air 260 Ultra	13 to 196 (0.2 to 3)	5
Chryso	98128	Chryso Air G 100	13 to 488 (0.2 to 7.5)	6
Chryso	98128	Chryso Air G 200	7 to 488 (0.1 to 7.5)	10
Chryso	98128	Chryso AirMax LDC	13 to 196 (0.2 to 3.0)	7
Chryso	98128	Chryso Air NVR	33 to 130 (0.5 to 2)	13

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Chryso	98128	Chryso Air TX	33 to 130 (0.5 to 2)	10
Chryso	98128	Daravair AT30	15 to 200 (0.25 to 3)	10
Chryso	98128	Daravair AT60	15 to 200 (0.25 to 3)	20
Chryso	98128	Daravair-1000	33 to 195 (0.5 to 3)	5
Chryso	98128	Darex AEA ED	30 to 302 (0.5to 5)	8
Chryso	98128	Darex II AEA	33 to 195 (0.5 to 5)	10
DarCole Products, Inc.	98387	DNF 842	16 to 260 (0.25 to 4)	9
DarCole Products, Inc.	98387	DSA 110	16 to 130 (0.25 to 2)	8
DarCole Products, Inc.	98387	DSA 112	16 to 260 (0.25 to 4)	7
Euclid Chemical Corp.	99924	Eucon AEA 92	6 to 260 (0.1 to 4)	6
Euclid Chemical Corp.	99924	Eucon AEA 92S	33 to 130 (0.5 to 2)	3
Euclid Chemical Corp.	99924	Eucon Air 30	33 to 260 (0.5 to 4)	3
Euclid Chemical Corp.	99924	Eucon Air 40	16 to 260 (0.25 to 4)	13
Euclid Chemical Corp.	99924	Eucon Air Mac-12	6 to 260 (0.1 to 4)	14
Euclid Chemical Corp.	99924	Eucon Air MAC6	6 to 260 (0.1 to 4)	7
Euclid Chemical Corp.	99924	Eucon Air Mix	33 to 65 (0.5 to 1)	15
Euclid Chemical Corp.	99924	Eucon Air Mix 200	33 to 65 (0.5 to 1)	15
Cemex Admix USA	98358	Isosphere 2004	6 to 320 (0.1 to 5)	5

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Cemex Admix USA	98358	Isosphere 5004	6 to 320 (0.1 to 5)	4
Cemex Admix USA	98358	Isosphere 9004	6 to 320 (0.1 to 5)	4
Cemex Admix USA	98358	Isosphere 9014	6 to 320 (0.1 to 5)	2
MAPEI Corp.	99828	Mapeair N	42 to 195 (0.65 to 3)	6
MAPEI Corp.	99828	Mapeair SA	8 to 130 (0.125 to 2)	7
MAPEI Corp.	99828	Mapeair SA 50	30 to 200 (0.5 to 30)	3
MAPEI Corp.	99828	Mapeair VR	8 to 195 (0.125 to 3)	15
Master Builders Solutions US LLC	99470	MasterAir AE 90	16 to 260 (0.25 to 4)	6
Master Builders Solutions US LLC	99470	MasterAir AE 200	8 to 260 (0.125 to 4)	12
Master Builders Solutions US LLC	99470	MasterAir AE 400	16 to 260 (0.25 to 4)	8
Master Builders Solutions US LLC	99470	MasterAir VR 10	16 to 260 (0.25 to 4)	13
Master Builders Solutions US LLC	99470	MasterCell 25	32 to 180 (0.5 to 2.75)	97
Chryso	98128	Quantec PL-466	130 to 325 (2 to 5)	13
RussTech	99379	RAE-260	33 to 130 (0.5 to 2)	9
RussTech	99379	RSA-10	33 to 130 (0.5 to 2)	7
RussTech	99379	RVR-15	33 to 130 (0.5 to 2)	15
Sika Corp.	99572	Sika AEA-14	65 to 195 (1 to 3)	8
Sika Corp.	99572	Sika Air	16 to 195 (0.25 to 3)	6

Producer	Producer Code	Product	Dosage in ml per 100 kg (oz per 100 lb. cement)	Approx. % Solids (for mix design)
Sika Corp.	99572	Sika Air 260	7 to 400 (0.1 to 6)	10
Sika Corp.	99572	Sika Air 360	7 to 400 (0.1 to 6)	7
Sika Corp.	99572	SikaControl AIR-160	7 to 400 (0.1 to 6)	3
Sika Corp.	99572	Sika Multi Air 25	13 to 195 (0.2 to 3)	6
Fritz-Pak Corp.	99803	Super Air Plus	16 to 81 (0.25 to 1.25)	97
Chryso	98128	Terapave	30 to 200 (0.5 to3)	10